

Xerox
WorkCentre PE220
23.60

600x600
150/50
6
5000/3000
600x600
4800x4800
Flatbed and Sheetfed
24
4/30
TWAIN, WIA

4
4
99

6
2
4/4

USB, Parallel
10.4
43.8x37.4x36.8
10
4

4
3.5/3.5

SPL
3.75
Windows, Linux, Mac
above 10.3

6/4
ABBYY FineReader 6,
PaperPort
31.08

11
11/14
11/15.5
3.5

13.6
10.5

19/209
22
1
4/9
7.12
6,000
26,500
61.80



+ Good print quality
- Expensive

While USB is the norm today, a parallel port interface is important when connecting to the old PCs that many offices still use.

One really good thing about the Canon MF3110 is its huge 64 MB of memory, which makes sure heavy documents are printed without using the PC's hard drive as the spooler, which slows down the process. The other printers are not in the league of the Canon MF3110 when it comes to memory; the closest any came was the Lexmark X215, with 16 MB.

Drivers, Manuals, OS Support

All the MFDs come with the drivers you'll need, as well as Quick Start Guides. Manuals in PDF format were also provided on the CDs that came bundled.

While Lexmark and Canon support only Windows, Xerox supports Windows as well as Linux and Macintosh.

Printing Languages

Almost all the MFDs, with the exception of the Lexmark, support the SPL (Samsung Printer Language) or CARPS (Canon Advanced Raster Printing System) printing languages. When the printer part of the MFD supports a printing language, it harnesses the power of the computer during printing; hence, the more powerful the computer, the better the printing performance will be.

OCR Applications

MFDs often come with an OCR application bundled. Optical Character Recognition (OCR) software is used to scan a printed text document and convert it to text formats understandable to the PC, such as MS Word or PDF. The Xerox WorkCentre PE220 comes with ABBYY FineReader 6.0, which is one of the best OCR software available. The Canon MF3110 comes with OmniPage, and the Samsung SCX-4521F comes with ReadIris Pro 9.

PERFORMANCE

Printing Warm-up Time

The warm-up time of an MFD indicates how soon it gets ready to print after it is switched on. Every time the printer goes into power saving mode, it will take some time to warm up, and that's why the warm up time is generally considered an important parameter.

The Canon MF 3110 is the quickest at warming up—it takes just six seconds, way better than the rest. The Samsung SF-565P and the Lexmark X215 are the slowest—they take 22 and 20 seconds respectively.

Print Quality And Speeds

In the first text print test, most of the MFDs fared almost equally well. The slowest was the Lexmark X215. The results of the combi-document test, too, were almost the same across all the MFDs.

Where print quality is concerned, the Samsung SCX-4521F and Xerox WorkCentre

PE220 were found to produce crisp prints. This was also evident in the combi-document print-outs, where these scored the highest in the concentric circles patterns test. In the minimum readable text test, the Lexmark X215 and Samsung SCX-4521F were the low scorers: they were only able to print 5-point text legibly at normal quality.

At best quality, almost all the MFDs printed 3-point text legibly, except for the Canon MF3110, which could manage only 4-point text. In general, most of the MFDs in this test are capable of excellent quality prints when set to best quality.

Scanning Speeds

Taking just eight seconds to scan our text document, the Samsung SCX-4521F emerged as the fastest scanner. The Samsung SF-565P logged the lowest average preview scan times of just 10 seconds. This was, of course, because of its lower dpi and also because it scans only in monochrome.



Samsung SCX-4100

The Canon MF 3110 was the fastest of the lot in our photo test—it clocked an amazing 61.5 seconds to scan our photograph at 600 dpi. The next-best timing was a distant 209 seconds by the Xerox WorkCentre PE220. The Lexmark X215 and Samsung SCX-4100 were the slowest, at 407 and 408 seconds respectively. We couldn't include the Samsung SF-565P in our photo test, since it only supports monochrome mode.

Quality

When it came to quality, the Canon MF 3110 and the Xerox PE220 were neck-to-neck at the top, with almost equal scores in tests such as the IT8 card test. The Lexmark X215 scored the lowest in the IT8 card test, and it was able to distinguish only 17 of the 22 boxes in the tonal deviation test.

OCR

Most of the MFDs fared well in our OCR test, which measures the degree to which the scanner part of the MFD can distinguish dark areas from light, which is what is required if the OCR software is to do its job well. The